CGGLNet: Semantic Segmentation Network for Remote Sensing Images Based on Category-Guided Global–Local Feature Interaction

Yue Ni, Jiahang Liu, *Member, IEEE*, Weijian Chi, Xiaozhen Wang, and Deren Li

***Abstract*— As spatial resolution increases, the information conveyed by remote sensing images becomes more and more complex. Large-scale variation and highly discrete distribution of objects greatly increase the challenge of the semantic segmentation task for remote sensing images. Mainstream approaches usually use implicit attention mechanisms or transformer modules to achieve global context for good results. However, these approaches fail to explicitly extract intraobject consistency and interobject saliency features leading to unclear boundaries and incomplete structures. In this article, we propose a category-guided global–local feature interaction network (CGGLNet), which utilizes category information to guide the modeling of global contextual information. To better acquire global information, we proposed a category-guided supervised transformer module (CGSTM). This module guides the modeling of global contextual information by estimating the potential class information of pixels so that features of the same class are more aggregated and those of different classes are more easily distinguished. To enhance the representation of local detailed features of multiscale objects, we designed the adaptive local feature extraction module (ALFEM). By parallel connection of the CGSTM and the ALFEM, our network can extract rich global and local context information contained in the image. Meanwhile, the designed feature refinement segmentation head (FRSH) helps to reduce the semantic difference between deep and shallow features and realizes the full integration of different levels of information. Extensive ablation and comparison experiments on two public remote sensing datasets (ISPRS Vaihingen dataset and ISPRS Potsdam dataset) indicate that our proposed CGGLNet achieves superior performance compared to the state-of-the-art methods.**

***Index Terms*— Category-guided, global contextual information, remote sensing, semantic segmentation, transformer.**

I. Introduction

SEMANTIC segmentation of remote sensing images [1], [2], as a highlight topic in remote sensing image interpretation, focuses more on fine-grained classification labeling of each pixel in the image compared to the classification task [3] and has been widely used in scenes such as road extraction [4], [5], disaster monitoring [6], and land cover [7]. However, as the imaging resolution continues increasing, the information conveyed becomes more complex. Traditional machine learning algorithms [8], such as support vector machines [9] and conditional random fields [10], gradually find it difficult to meet the demands of high-resolution remote sensing image applications.

With the booming development of deep learning technology [11], researchers have applied convolutional neural network (CNN)-based methods to image segmentation tasks. Full convolutional networks (FCNs) [12] introduced CNNs into the field of image segmentation for the first time and achieved satisfactory segmentation accuracy. However, the image spatial location information was ignored during the segmentation process, resulting in insufficiently fine segmentation results. Immediately after that, the symmetric encoder and decoder structure of U-Net [13] successfully compensate for the lost detail information through skip connections, which further improves the segmentation accuracy. Nevertheless, the skip connection fails to effectively distinguish and fuse the underlying and high-level features, which limits the feature expression capability of the network. Inspired by the above networks, many researchers have recognized the importance of multiscale features in the semantic segmentation of remote sensing images [14], [15]. The DeepLab series of networks [16], [17], [18] has improved segmentation accuracy by expanding the model sensing field without increasing the parameters through the design of the atrous spatial pyramid pooling (ASPP) module. Pyramid scene parseing network (PSPNet) [19] cleverly applies adaptive global pooling to construct the pyramid pooling module (PPM), which captures features of different scales and fuses them to improve the segmentation accuracy.

However, distinguishing from natural images, the distribution of objects in remote sensing images is more discrete, as shown in Fig. 1. Relying only on local context information for pixel classification may lead to blurred edges between classes and incomplete segmented objects. Effectively extracting global context information, on the other hand, can more accurately characterize pixel points. Therefore, many researchers have further improved the segmentation accuracy by introducing the attention mechanism [20], [21], [22], [23], [24] or the transformer module [25], [26], [27], [28] in the framework to model the global context information, which fully proves that the effective fusion of local and global

Manuscript received 20 January 2024; revised 28 February 2024; accepted 14 March 2024. Date of publication 20 March 2024; date of current version 1 April 2024. This work was supported in part by the Innovative Talent Program of Jiangsu under Grant JSSCR2021501 and in part by the National Science and Technology Major Project (TW-2). *(Corresponding author: Jiahang Liu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3379398

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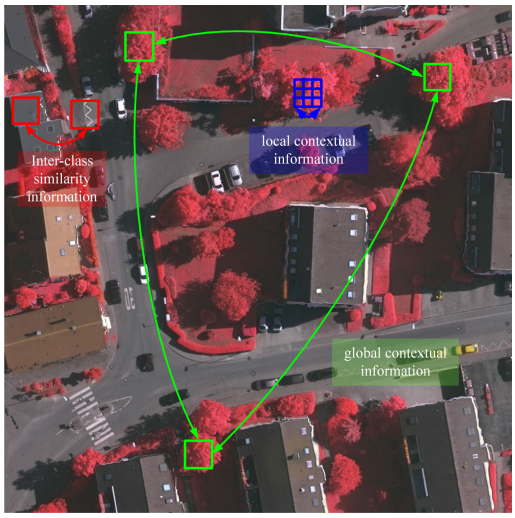


Fig. 1. Illustration of global and local contextual information. Blue blocks indicate local contextual information, green blocks/red blocks indicate intraclass/interclass local contextual information, and green/red arrows indicate intraclass/interclass global contextual information interactions.

features helps to characterize the classes of pixel points more accurately.

Currently, most of the networks use the attention mechanism or transformer module to model the global context information by generating multichannel features in the backbone network and performing matrix multiplication computation to obtain the correlation weight matrix [29], [30]. As shown in Fig. 1, the red boxes indicate local regions belonging to two different types of objects that have extremely similar local features. However, when using the above method for global context modeling, the locally similar features of different objects interact with each other, resulting in poor differentiation of the extracted features, which causes feature aliasing and, consequently, unclear segmentation boundaries and incomplete structures. In addition, the computational cost of this method is relatively high. In fact, for appropriately sized feature maps, we can still perform some supervised constraints by using truth values to generate the correlation matrices associated with the classes. In this way, intraclass consistency features and interclass differentiation features can be extracted.

Based on this consideration, we proposed a network framework for modeling global contextual information using category information-guided global contextual information, called category-guided global–local feature interaction network (CGGLNet). The CGGLNet contains two main modules, the global–local supervised transformer module (GLSTM), which includes the category-guided supervised transformer module (CGSTM), and the adaptive local feature extraction module (ALFEM). These two modules are used to effectively capture global and local contextual information. The CGSTM guides the modeling of global contextual information by estimating the potential category of each pixel point in the spatial dimension, thus improving the modeling capability of global information. The ALFEM employs a gate mechanism to adaptively capture multiscale local detailed features, thus improving the modeling capability of local contextual information. In the output part, a feature refinement segmentation head (FRSH) is also introduced to narrow the semantic gap between shallow and deep features to ensure that different features are fully fused. The main contributions can be summarized as follows.

1) We propose a new semantic segmentation network, CGGLNet, which aims to construct segmentation models by channeling global contextual information through category information, extracting salient features that can effectively express intraobject consistency and interobject variability, and thus producing clearer segmentation boundaries and more complete structures.

2) We propose a GLSTM module (consisting of CGSTM and ALFEM in parallel) that can efficiently acquire global contextual information and perform multiscale local feature extraction on images to enrich the feature representation. CGSTM utilizes the coarse segmentation results to guide the extraction of global context information, obtaining global features that are more consistent with the same kind of features, and more distinguishable between different categories of features. ALFEM adaptively captures local detailed features of multiscale objects in remote sensing images by introducing a gate mechanism.

3) We propose an FRSH module for fully integrating deep semantic information and shallow spatial information from coarse to fine, narrowing the semantic gaps between features, and improving the expressive power of features. FRSH effectively captures the correlation between different channels, the correlation between different regions, and the long-range orientation information in the image through channel attention, spatial attention, and atrous convolution.

II. Related Work

A. CNN-Based Semantic Segmentation Methods

Since the introduction of FCNs [12], CNN-based methods have dominated the field of semantic segmentation in remote sensing. However, the design of the FCN decoder ignores the spatial information of 2-D images, which leads to a serious sense of fragmentation in the final segmentation result. To solve this problem, U-Net [13] introduces an encoder–decoder symmetric structure while compensating the spatial information of the high-level features in the decoder by skip connections, which further improves the segmentation accuracy. However, the fixed-size convolutional kernel limits the network's feature extraction capability for large-scale objects.

To improve the network's ability to extract multiscale features, researchers have mostly used atrous convolution and adaptive pooling for multiscale feature extraction [31], [32]. The DeepLab series of networks [16], [17], [18] expands the model sensing domain and improves the segmentation accuracy without adding too many parameters through the design of the ASPP module. PSPNet [19] cleverly applies adaptive global pooling to build the PPM to capture features at different scales and fuse them to improve segmentation accuracy.

However, with the increasing resolution of remote sensing images, the object's local similarity is high, so global contextual information is needed to complement it. By incorporating the attention mechanism into the network [33], [34], [35], [36] architecture for global context information extraction, the researchers get rid of the limitation of focusing on

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the local area [22]. Multi-stage attention ResU-Net (MAResU-Net) [20] greatly reduces the computational cost by the designed linear attention mechanism (LAM) and applies it to the multiscale feature maps for global context extraction. Based on this, multi-attention network (MANet) [21] performs the integration of global dependencies in channel and spatial dimensions through the designed efficient linear attention module. Synergistic attention perception neural network (SAPNet) [23] performs the modeling of interchannel dependencies while retaining rich spatial details through the designed synergetic attention module (SAM). Supervised long-range correlation network (SLCNet) [24] supervises the self-attention mechanism by constraining the self-attention weights for full interaction with the same class of object features, which in turn improves the model's ability to model global contextual information. Although the embedding of the attention mechanism improves the feature representation of the network to some extent, the over-reliance on convolutional operations within these modules limits the ability to characterize global features.

In this article, we propose an ALFEM, which aims to enhance the adaptive processing capability for various-sized targets in multiscale images. ALFEM enhances the network's effective local feature extraction for different-sized targets by employing the weight vectors generated by the channel attention mechanism to select features generated by different-sized convolutional kernels. Meanwhile, an FRSH is proposed to narrow the semantic gap between the fused features by effectively fusing the channel attention, spatial attention, and directional atrous convolution to further improve the feature expression capability and network segmentation accuracy [27], [28].

B. Transformer-Based Semantic Segmentation Methods

In recent years, with the rise of transformer [37] in natural language processing (NLP) and computer vision tasks, researchers have introduced it to semantic segmentation tasks [38], [39]. Transformer-based semantic segmentation methods can be broadly classified into two categories. The first category is the architecture where the encoder–decoder is composed of transformers, and typical networks include Segmenter [40], SegFormer [41], and SwinUNet [42]. The second category is the hybrid framework combining CNN and transformer. For the task of semantic segmentation of remote sensing images, the local information present in the image cannot be ignored. Therefore, most of the frameworks of transformer-based semantic segmentation methods for remote sensing images are based on the second category [43], [44], [45]. DC-Swin [25] complements each other by utilizing densely connected feature aggregation modules (DCFAMs) to facilitate the fusion of multiscale global features. Combining swin transformer with UNet (CST-UNet) [26] fuses both local and global features by utilizing a dual encoder architecture with a backbone network that extracts both local and global features simultaneously and fuses these two features through a position-aware attention fusion module (PAFM). AerialFormer [46] utilizes multidilated CNNs (MD-CNNs) to aggregate multiscale features from the output of the transformer encoder, which enriches the global and local contextual representation of the network and improves the segmentation accuracy. Due to the large size of the original remote sensing image data, training using a pure transformer architecture takes a lot of time. Therefore, the lightweight design for the transformer module is also a hot research topic. To improve the modeling efficiency of the global context of the transformer, UNetFormer [27] proposed a global–local transformer block (GLTB) by using a pooling operation consistent with the window size to establish the information association between windows while preserving the local detail features. CNN and multiscale transformer fusion network (CMTFNet) [28] reduces the size of the feature maps by using an adaptive pooling operation, which reduces the global context information modeling computation while extracting multiscale global context information.

In this article, we propose CGSTM to enhance the interaction of global contextual information of the network. CGSTM replaces query tensors ($\boldsymbol{Q}$), and key tensors ($\boldsymbol{K}$) in the traditional multihead transformer module with coarser segmentation results of lower channel dimensionality and obtains category-strongly correlated weight maps with lower computational effort. Moreover, each attention head can fully extract intraclass consistency features and interclass saliency features in the global view. Therefore, this approach helps the network to produce more consistent segmentation results for pixels or regions belonging to the same class at different locations in the global view.

III. Proposed Method

In this section, we describe the architecture and constituent modules of CGGLNet in detail. In Section III-A, the general architecture of the network is described. The GLSTM for extracting global and local context information will be divided into Sections III-B1 and III-B2, which describe the CGSTM and the ALFEM, respectively. In Section III-C, we will discuss in detail the FRSH used to reduce the gap between deep and shallow features. Finally, in Section III-D, the loss function for updating the network parameters is presented.

A. Network Architecture

The overall architecture of CGGLNet is shown in Fig. 2, which adopts a classical encoder–decoder structure with a skip connection. In the encoder part, we choose to use ResNet-50 with pretrained weights as the backbone network for multiscale feature extraction. In the decoder part, first, the channel number of the four deepest feature maps is adjusted to 192 by 1×1 convolution operation. Then, we introduce the feature fusion module (FSM) module to selectively aggregate the GLSTM-processed features of the previous layer and the channel-unified features of the current layer to enhance the feature representation. Meanwhile, we pass the fusion features of the FSM module to the auxiliary output module (AOM) to generate a rough segmentation result of the current scale features, which is input to the GLSTM in the same layer to assist in global feature interaction. Note that at the deepest feature maps, given the weak spatiality of its features, we chose to forgo the use of GLSTM for feature optimization and instead use the unsupervised global–local transformer module (GLTM) for global and local feature optimization.

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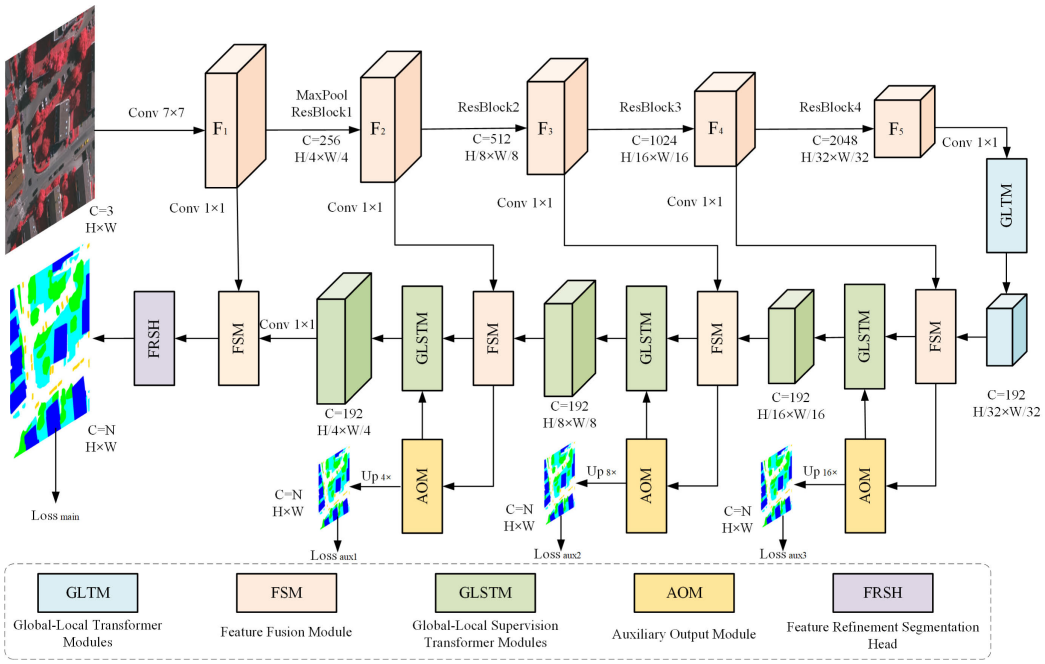


Fig. 2. Illustration of the network architecture of CGGLNet. Multiscale features $\boldsymbol{F}_1$–$\boldsymbol{F}_5$ are generated by ResNet-50. C is the number of channels, N is the number of classification categories, H and W represent the height and width of the input image, respectively, and Up represents bilinear interpolation upsampling.

Finally, the aggregated features output from GLSTM are reduced to 64 channels by 1 × 1 convolution, and after FSM fusion with the shallowest features, they are inputted into the FRSH module to generate the final segmentation result map. We update the network parameters by simultaneously constraining the loss function on the final segmentation result and the coarse segmentation result output from AOM. The weighting and operational expressions of FSM for features are as follows:

$$\boldsymbol{F}_{\mathrm{FSM}} = \alpha \cdot \boldsymbol{F} + (1-\alpha) \cdot Up_{2\times}\left(\boldsymbol{F}_{\mathrm{GL(S)TM}}\right) \tag{1}$$

where $\boldsymbol{F}$ denotes the features generated by the ResNet backbone, $\boldsymbol{F}_{\mathrm{GL(S)TM}}$ denotes the output features after the GLTM or GLSTM module, $Up_{2\times}(\cdot)$ denotes the bilinear interpolation twice upsampling, α denotes the weight parameters learned by the network, and $\boldsymbol{F}_{\mathrm{FSM}}$ denotes the output fusion features of the FSM. The coarse segmentation of the fused features is performed using the AOM module, which is computed in the following equation:

$$\boldsymbol{aux} = \mathrm{Conv}_{1\times1}(\mathrm{Drop}(\mathrm{CBR}_{3\times3}(\boldsymbol{F}_{\mathrm{FSM}}))) \tag{2}$$

where $\boldsymbol{F}_{\mathrm{FSM}}$ is the output of the FSM; $\mathrm{CBR}_{3\times3}(\cdot)$ is a 3 × 3 convolutional kernel, a batch normalization layer, and a rectified linear unit (ReLU) activation layer; $Drop(\cdot)$ is a regularization method set to 0.1; $\mathrm{Conv}_{1\times1}(\cdot)$ is a 1 × 1 convolution; and $\boldsymbol{aux}$ is the result of coarse segmentation with N (represents the number of categories) channels.

B. Global–Local Supervised Transformer Module

One of the challenges in semantic segmentation of remotely sensed images lies in the fact that as the resolution increases, the detailed texture features of many objects become richer, which provides neural networks with the opportunity to learn more features. However, this also brings the risk of feature overlap between objects. Meanwhile, remote sensing images have a wider geographic area coverage than natural images, which means that objects of interest may appear at any location in the image at different scales. Pixel features of the same class exhibit high correlation, independent of spatial location. Therefore, we need to effectively model the global features of an image while focusing on the multiscale local features of the image to obtain features with intraclass consistency and interclass differentiation.

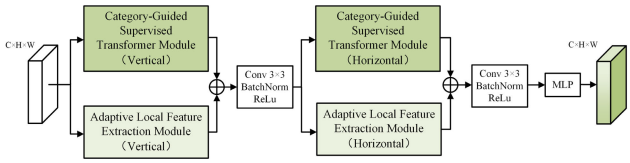


Fig. 3. Details of GLSTM. GLSTM performs category-guided global feature interactions and adaptive local feature interactions in parallel for the input feature maps in vertical and horizontal directions in turn and outputs the final features after MLP.

To address these challenges, we propose the GLSTM with the structure shown in Fig. 3. To extract features with directionality and at the same time reduce the computational cost. We adopt the strategy of performing global–local feature extraction in the vertical direction first, followed by global–local feature extraction in the horizontal direction [47], [48]. After multilayer perceptron (MLP) processing, the optimized features are obtained.

1) Category-Guided Supervised Transformer Module: The core of global feature modeling by transformer lies in the generation of the weight matrix, which is obtained from the

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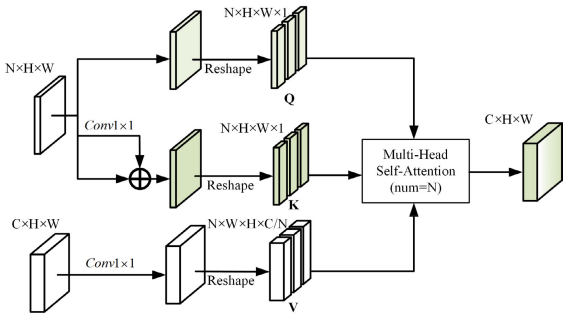


Fig. 4. Details of CGSTM. Taking the vertical direction as an example, the coarsely segmented results are processed to obtain $\boldsymbol{Q}$ and $\boldsymbol{K}$. The multidimensional features are processed accordingly to obtain $\boldsymbol{V}$. Next, they are fed into a self-attention mechanism consisting of N attention heads to obtain the global contextual information guided by the categories.

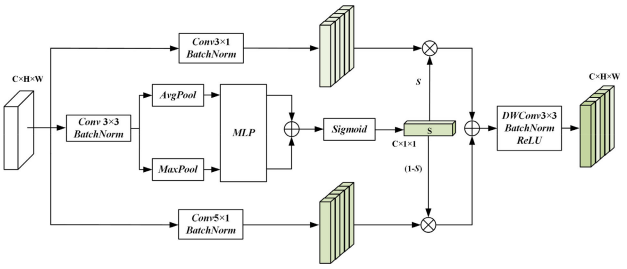


Fig. 5. Details of ALFEM. Taking the vertical direction as an example, the multidimensional features are used to adaptively select small-scale vertical direction local features and large-scale vertical direction local features by generating channel selection weights through feature selection gate branches to realize adaptive capture of local features of multiscale objects.

product of the query tensor ($\boldsymbol{Q}$) and the key tensor ($\boldsymbol{K}$). Typically, $\boldsymbol{Q}$ and $\boldsymbol{K}$ are obtained by performing a fully connected operation on multichannel features. However, the number of channels of the multichannel features is usually large, and thus, more computational cost may be introduced in computing the product of $\boldsymbol{Q}$ and $\boldsymbol{K}$. In addition, the weight maps obtained by self-supervised learning of the network may not be high enough in terms of interpretability. In contrast, the original intention of using transformer is to learn the commonalities of features of the same class of pixel points and the differences of features of different classes of pixel points from a global perspective.

To address the above problems, we propose the CGSTM, the detailed structure of which is shown in Fig. 4. This module reduces the computational cost of generating the weight matrix by replacing $\boldsymbol{Q}$ and $\boldsymbol{K}$ in the multihead transformer with the coarse segmentation results predicted by the same layer of features and reduces the channel dimensions of $\boldsymbol{Q}$ and $\boldsymbol{K}$ to N, where N is the number of segmented categories. Meanwhile, considering that the coarse segmentation results generated in each layer are not completely accurate, we use a 1×1 convolution to update $\boldsymbol{K}$ moderately, but at the same time, we want $\boldsymbol{K}$ to retain the complete coarse segmentation results, so we adopt the residual structure. The number of attentional heads of the multihead transformer matches the number of categories to be segmented, ensuring that each head focuses on a specific category.

Given the inputs of CGSTM as $\boldsymbol{aux} \in \mathbb{R}^{N\times H\times W}$, $\boldsymbol{F}_{FSM} \in \mathbb{R}^{C\times H\times W}$. Keeping the original $\boldsymbol{aux}$ as $\boldsymbol{Q}$, map $\boldsymbol{aux}$ and $\boldsymbol{F}_{\mathrm{FSM}}$ to two different feature vectors, named $\boldsymbol{K} \in \mathbb{R}^{N\times H\times W}$ and $\boldsymbol{V} \in \mathbb{R}^{C\times H\times W}$, respectively, using two trainable linear projections, which operate as follows:

$$(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \left(\boldsymbol{aux}, \boldsymbol{aux}\boldsymbol{W}^k + \boldsymbol{aux}, \boldsymbol{F}_{FSM}\boldsymbol{W}^v\right) \tag{3}$$

where $\boldsymbol{W}^k$ and $\boldsymbol{W}^v$ are the weight matrices used to generate the linear transformation of the key tensor and value tensor, respectively. As mentioned above, multiple self-attention is performed in vertical and horizontal directions, respectively, rather than on the entire image. Taking the vertical direction as an example, the above three tensors are reshaped into $\boldsymbol{Q} \in \mathbb{R}^{N\times W\times H\times 1}$, $\boldsymbol{K} \in \mathbb{R}^{N\times W\times H\times 1}$, and $\boldsymbol{V} \in \mathbb{R}^{N\times W\times H\times C/N}$ and assigned to each attention head. Here, N represents the number of categories and also the number of attention heads. Calculate the self-attention feature with the following formula:

$$\mathrm{Attention}_v^k = \mathrm{Softmax}\left(\frac{Q_v^k \times K_v^{k^T}}{\sqrt{d_{K_v^k}}}\right) \times V_v^k \tag{4}$$

$$F_{\mathrm{CGSTM}_v} = \mathrm{concat}\left(\mathrm{Attention}_v^1, \mathrm{Attention}_v^2, \ldots, \mathrm{Attention}_v^N\right) \tag{5}$$

where $\boldsymbol{Q}_v^k \in \mathbb{R}^{W\times H\times 1}$, $\boldsymbol{K}_v^k \in \mathbb{R}^{W\times H\times 1}$, and $\boldsymbol{V}_v^k \in \mathbb{R}^{W\times H\times C/N}$ denote the inputs of the attention head numbered k, v represents the vertical direction, $d_{K_v^k}$ represents the number of channels of $\boldsymbol{K}_v^k$, and $\mathrm{Attention}_v^k$ denotes the output of the attention head numbered k. Channel splicing is performed by concat$(\cdot)$ to splice the features of N attention heads together to form the final vertical global optimization feature, denoted by F_{CGSTM_v}.

The weight values obtained after the multiplication of $\boldsymbol{Q}$ and $\boldsymbol{K}$ by Softmax can reflect the correlation of two points in the spatial dimension. If this value is larger, the category of the two points in the spatial dimension of the response is consistent. For the horizontal direction, the vectors in (3) are reshaped into $\boldsymbol{Q} \in \mathbb{R}^{N\times H\times W\times 1}$, $\boldsymbol{K} \in \mathbb{R}^{N\times H\times W\times 1}$, and $\boldsymbol{V} \in \mathbb{R}^{N\times H\times W\times C/N}$, and (4) and (5) are used to model the global context information in the horizontal direction. For global context information modeling of feature maps in vertical and horizontal directions sequentially, the category-guided global context-based features are obtained.

2) Adaptive Local Feature Extraction Module: For remote sensing images, relying only on global features cannot adequately capture the details and local features in the images because remote sensing images usually contain complex feature structures and rich texture information. In the task of semantic segmentation of remote sensing images, the introduction of local features becomes crucial. By combining global and local features, we can decipher the complex structures and feature distribution in images more comprehensively.

Based on the aforementioned ideas, we constructed an ALFEM, which aims to be able to adaptively capture the local features of multiscale objects to improve the overall feature description capability. The specific structure of this module is shown in Fig. 5. By using a smaller size convolutional kernel, we can realize the fine extraction of local features of small-size objects, while using a larger size convolutional kernel can better capture the local features of large-size

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objects. Meanwhile, inspired by squeeze-and-excitation network (SE-Net), we introduce a gate mechanism based on channel attention to quantify the affinity of each channel feature to objects at different scales. Unlike traditional SE modules, we combine the results of maximum pooling and average pooling to determine the weights more accurately while utilizing 1×1 convolution instead of linear layers to make it more adaptable to 2-D images. This fusion strategy aims to capture the importance of channel features more comprehensively. To harmonize with the CGSTM strategy, we similarly introduce adaptive local feature extraction for the entire image in both vertical and horizontal directions. This design ensures that the module can effectively capture the local information of the image in multiple directions.

Specifically, ALFEM is divided into three parallel branches: small-scale local feature extraction, large-scale feature extraction, and feature selection gate weights. Similarly, we take the vertical direction as an example. First, the input feature $\boldsymbol{F}_{\mathrm{FSM}}$ is passed through a 3×1 convolutional layer, a batch normalization layer, a 5×1 convolutional layer, and a batch normalization layer to generate vertically oriented small-scale local feature $\boldsymbol{F}_{\mathrm{samll_}v}$ and vertically oriented large-scale local feature $\boldsymbol{F}_{\mathrm{large_}v}$ describing the local features of small- and large-size objects, respectively. Meanwhile, the input features $\boldsymbol{F}_{\mathrm{FSM}}$ are subjected to a channel attention mechanism to generate local feature selection switch weights $\boldsymbol{S} \in \mathbb{R}^{C\times 1\times 1}$, which represent the affinity of each channel for small-size objects. Conversely, $\mathbf{1} - \boldsymbol{S}$ represents the affinity of each channel for large-size objects. Summing $\boldsymbol{S}$ and $\mathbf{1} - \boldsymbol{S}$ by multiplying them with the generated $\boldsymbol{F}_{\mathrm{samll_}v}$ and $\boldsymbol{F}_{\mathrm{large_}v}$, respectively, yields the locally enhanced features $\boldsymbol{F}_{\mathrm{local_}v}$ in the vertical direction. In summary, the above process can be expressed as follows:

$$\boldsymbol{F}_{\mathrm{small_}v} = \mathrm{CB}_{3\times 1}(\boldsymbol{F}_{\mathrm{FSM}}),\ \boldsymbol{F}_{\mathrm{large_}v} = \mathrm{CB}_{5\times 1}(\boldsymbol{F}_{\mathrm{FSM}}) \tag{6}$$

$$\begin{aligned}\boldsymbol{S} = \mathrm{sigmoid}(&\mathrm{MLP}(\mathrm{AvgPool}(\mathrm{CB}_{3\times 3}(\boldsymbol{F}_{\mathrm{FSM}}))) \\ &+ \mathrm{MLP}(\mathrm{MaxPool}(C\,B_{3\times 3}(\boldsymbol{F}_{\mathrm{FSM}}))))\end{aligned} \tag{7}$$

$$\boldsymbol{F}_{\mathrm{local_}v} = \boldsymbol{S} \cdot \boldsymbol{F}_{\mathrm{small_}v} + (\mathbf{1} - \boldsymbol{S}) \cdot \boldsymbol{F}_{\mathrm{large_}v} \tag{8}$$

where $C\,B(\cdot)$ is a convolutional kernel and batch normalization layer, sigmoid$(\cdot)$ is the activation function, AvgPool$(\cdot)$ and MaxPool$(\cdot)$ denote average pooling and maximum pooling, respectively, and MLP$(\cdot)$ denotes the fully connected layer consisting of 1×1 convolutional kernels.

The global features $F_{\mathrm{CGSTM_}v}$ in the vertical direction are further integrated to form the global–local context. Subsequently, the global–local context in the vertical direction is obtained by performing 3×3 convolution, batch normalization operations, and applying the ReLU activation function. Such a treatment allows for a better characterization of the fine-grained global–local context in the vertical direction.

C. Feature Refinement Segmentation Head

The twice-downsampled feature maps generated by ResNet-50 contain rich detailed texture information and some noise details, such as some edge information inside large-size targets. In contrast, the bottommost feature map contains rich deep semantic information through layer-by-layer upsampling. Direct splicing or summing does not fully utilize these two features. Therefore, to effectively fuse these two features, reduce the semantic difference between the two features, and refine the feature representation, we introduce the FRSH, whose structure is shown in Fig. 6.

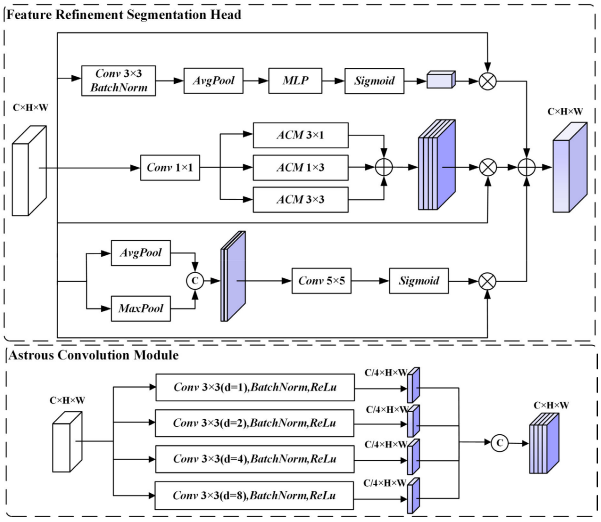


Fig. 6. Details of FRSH. FRSH performs channel attention, spatial attention, and multiscale enhancement operations for the features that have been initially fused by the FSM to further minimize the semantic differences between shallow and deep features.

We take the output of the shallowest FSM as the input to the FRSH. Subsequently, for the initial fused features, we perform three enhancement paths: channel enhancement, spatial enhancement, and multiscale enhancement. Specifically, the channel enhancement path uses global pooling to generate a channel's attention weight ($\boldsymbol{attention}_c \in \mathbb{R}^{C\times 1\times 1}$), which is multiplied by the input features to obtain the channel-enhanced fused features. The spatial enhancement path is similar to the convolutional block attention module (CBAM) [34] module in that the spatial attention of two channels is generated by maximum and average pooling, and then, the spatial attention weights ($\boldsymbol{attention}_s \in \mathbb{R}^{1\times H/2\times W/2}$) are generated by a 5×5 convolution and a Sigmoid for degradation and activation, which are multiplied with the input features to obtain the spatially enhanced fusion features. The multiscale enhancement path draws on our previously proposed atrous convolution module (ACM) [49], whose structure is shown in Fig. 6. Considering that the spatial arrangements and sizes of objects in remote sensing images from different scenes vary greatly, we introduce the ACM to generate multidirectional and multiscale attention, which is multiplied with the input features to obtain multiscale enhanced fusion features. Finally, the summation operation is performed on the results of the three branches to obtain the final fusion features, and channel compression and upsampling are performed to obtain the final segmentation results.

D. Loss Function

For our proposed CGGLNet, we supervise the final segmentation results output by FRSH and the three coarse segmentation results generated by AOM separately by using a loss function to train the whole network. Meanwhile, the coarse segmentation results generated by AOM are used to guide CGSTM for modeling global information, and their

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results directly affect the correctness of global context information modeling. Therefore, the final segmentation results have the same loss function as the coarse segmentation results. Equal weighted summation of cross-entropy loss (L_{CE}) and Dice loss (L_{Dice}) is employed to combine their advantages and balance the classification accuracy for each pixel and the concern for spatial overlap. This helps to improve the performance of the model in semantic segmentation tasks. Specifically, the expressions for L_{CE} and L_{Dice}, respectively, are as follows:

$$L_{ce} = -\frac{1}{\mathrm{HW}} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{n=1}^{N} \left(y_{ij,n} \log \hat{y}_{ij,n} \right) \tag{9}$$

$$L_{\mathrm{dice}} = 1 - \frac{2|\hat{y} \cap y|}{|\hat{y}| + |y|} \tag{10}$$

where $\hat{y}$ denotes the network prediction segmentation result and y denotes the ground truth. N denotes the number of categories, The subscript ij, n denotes the ith row and jth column pixel in the nth segmentation result map.

To better combine the main loss L_{main} and the auxiliary loss L_{aux} and L_{aux} are multiplied by the coefficient λ. Thus, the total loss can be expressed as

$$L = L_{\mathrm{main}} + \lambda (L_{\mathrm{aux1}} + L_{\mathrm{aux2}} + L_{\mathrm{aux3}}) \tag{11}$$

where λ is a hyperparameter, which is used to balance the influence of the two loss functions. Through the ablation experiment, λ is finally set to 1.0. Both L_{main} and L_{aux} are equal to $L_{ce} + L_{\mathrm{Dice}}$, and L_{aux1}, L_{aux2}, and L_{aux3} are the coarse segmentation results of the three AOMs.

IV. Dataset and Experimental Setting

A. Datasets

1) *ISPRS Potsdam:* The Potsdam dataset contains 38 orthophotos with a ground sampling distance (GSD) of 5 cm and a size dimension of 6000 × 6000. Each image contains the four bands of near infrared (IR), red (R), green (G), and blue (B), as well as the corresponding digital surface model (DSM) and normalized digital surface model (NDSM) data. The entire image is segmented into six categories based on pixels: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. In the experiments in this article, we use only red, green, blue (RGB) image data for training and testing. Based on previous work, we used 23 images for training and 14 images for testing, and the IDs of the test images are 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13. The remaining images (except 7_10 which has labeling errors) were used for training. Each image was cropped into 1024 × 1024 image blocks.
2) *ISPRS Vaihingen:* The Vaihingen dataset contains 33 true orthophoto (TOP) image blocks of different sizes with a GSD of 9 cm, and each TOP image block consists of IR, R, and G channels and contains both DSM and NDSM data. Its classification categories are the same as the Potsdam dataset. In the experiments in this article, we only use TOP image blocks. We selected 16 images for training and 17 images for testing, and the IDs of the test images are 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38. The remaining images are used for training. Each image was similarly cropped into 1024 × 1024 image blocks.

B. Implementation Details

The experiments in this article are based on the Windows 10 operating system and are trained and validated on a single NVIDIA RTX 3090 graphics processing unit (GPU) using the PyTorch framework to ensure fair comparisons. In the training phase, we used the AdamW optimizer and loaded the pretraining weights of the encoder. Specifically, the initial learning rate of the encoder was set to $6e^{-5}$, the initial learning rate of the rest was $6e^{-4}$, and the overall weight decay parameter was 0.01. We also used a cosine strategy to tune the learning rate.

During network training, we performed multiple aspects of data enhancement, including random cropping into 512 × 512 image blocks, random horizontal flipping, random vertical flipping, and random multiscale [0.5×, 0.75×, 1×, 1.25×, 1.5×] scaling. For the Potsdam dataset, we set the training epoch to 100 and the batch size to 4; for the Vaihingen dataset, we set the training epoch to 45 and the batch size to 4.

C. Evaluation Metrics

To be able to quantitatively assess the effectiveness of the proposed model, we used three metrics: overall accuracy (OA), mean intersection over union (mIoU), and $F1$ score ($F1$). The formulas are as follows:

$$\mathrm{OA} = \frac{\sum_{n=1}^{N} \mathrm{TP}_n}{\sum_{n=1}^{N} TP_n + \mathrm{FP}_n + \mathrm{TN}_n + \mathrm{FN}_n} \tag{12}$$

$$\mathrm{mIoU} = \frac{1}{N} \frac{\sum_{n=1}^{N} \mathrm{TP}_n}{\sum_{n=1}^{N} \mathrm{TP}_n + \mathrm{FP}_n + \mathrm{FN}_n} \tag{13}$$

$$\mathrm{precision}_n = \frac{\mathrm{TP}_n}{\mathrm{TP}_n + \mathrm{FP}_n}, \mathrm{recall}_n = \frac{\mathrm{TP}_n}{\mathrm{TP}_n + \mathrm{FN}_n} \tag{14}$$

$$\mathrm{F1Score}_n = 2 \times \frac{\mathrm{precision}_n \times \mathrm{recall}_n}{\mathrm{precision}_n + \mathrm{recall}_n} \tag{15}$$

where TP_n, FP_n, TN_n, and FN_n denote true-positive counts, false-positive counts, true-negative counts, and false-negative counts, respectively, all of which are specifically related to the given object indexed as class n. Meanwhile, we evaluate the complexity of the network model by the number of floating point operations per second (FLOPS) and the number of parameters of the network, where the input data size is 3 × 1024 × 1024.

V. Experimental Results and Analysis

A. Ablation Study

1) Effectiveness of Each Module of CGGLNet: To quantitatively evaluate the effectiveness of each of our proposed modules, we conducted the corresponding ablation experiments on two datasets, the results of which are shown in Table I.

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TABLE I
QUANTITATIVE RESULTS OF INTRODUCING MODULES IN THE PROPOSED METHOD ON THE VAIHINGEN DATASETS

Datasets	Method	Params (M)	FLOPs (GFLOPs)	Mean F_1(%)	OA(%)	mIoU (%)
Vaihingen	Baseline	24.51	361	89.75	90.53	81.65
	Baseline+FRSH	24.62	388	90.24	90.77	82.41
	Baseline+ALFE	29.92	477	90.27	90.83	82.45
	Baseline+CGSTM	29.32	462	90.62	91.03	83.06
	Baseline+GLSTM	34.73	578	90.83	91.26	83.40
	Baseline+ GLSTM +FRSH	34.83	605	**91.12**	**91.47**	**83.89**
Potsdam	Baseline	24.51	361	91.54	90.08	84.70
	Baseline+FRSH	24.62	388	92.12	90.65	85.60
	Baseline+ALFE	29.92	477	92.08	90.42	85.51
	Baseline+CGSTM	29.32	462	92.40	90.78	86.11
	Baseline+GLSTM	34.73	578	92.76	91.37	86.73
	Baseline+ GLSTM +FRSH	34.83	605	**92.96**	**91.58**	**87.11**

*The optimal values in each column are in bold.

1) *Baseline:* The baseline network in this article takes the backbone network as the U-Net of ResNet-50 as a reference. In contrast to the traditional U-Net, the decoder first standardizes the channel numbers of the output multiscale features to 192 and utilizes FSM for feature fusion between adjacent layers. Meanwhile, the loss function for the network training employs an equal-weighted summation of the cross-entropy loss and the Dice loss to more fairly represent the effectiveness of the proposed module.
2) *GLSTM:* In our baseline model, we constructed **Baseline+ GLSTM** by embedding three GLSTM on shallow features and embedding GLTM on the deepest features to achieve effective extraction of global–local context information. To verify the effectiveness of the proposed global and local modules, we constructed **Baseline+ ALFE** by masking the global branches in the four modules and constructed **Baseline+ CGSTM** by masking the local branches in the modules to verify the effectiveness of ALFE and CGSTM, respectively. As shown in Table I, after introducing ALFE, the network's average $F1$, OA, and mIoU on the two datasets are improved by at least 0.52%, 0.30%, and 0.80%, respectively, validating the effectiveness of ALFE. The introduction of CGSTM improved the average $F1$, OA, and mIoU of the network on the two datasets by at least 0.86%, 0.50%, and 1.41%, respectively, validating the effectiveness of CGSTM. The combination of the two further improves the segmentation accuracy by at least 1.08%, 0.73%, and 1.75% for the average $F1$, OA, and mIoU on the two datasets, respectively, confirming the complementary nature of ALFE and CGSTM in terms of features. Although the embedding of GLSTM increases the number of parameters of the network by 10.22M and FLOPs by 217, its OA improvement is significant. To further demonstrate the advantages of the proposed GLSTM module, we replaced the GLSTM module in CGGLNet and constructed a variant of CGGLNet using other state-of-the-art global–local context modules for ablation studies. Although the use of the GLSTM module increases the complexity of the network, we achieve the highest mIoU on the Vaihingen dataset due to its category-guided global context information modeling and adaptive local context information extraction, as shown in Table II.
3) *FRSH:* We embed FRSH into **Baseline+ GLSTM** to form our CGGLNet (**Baseline+ GLSTM + FRSH**). Meanwhile, to validate the effectiveness of FRSH, we embedded it into the Baseline alone to construct **Baseline+ FRSH**. As shown in Table I, the Baseline embedded with FRSH improves the average $F1$, OA, and mIoU on both datasets by at least 0.49%, 0.24%, and 0.76%, respectively, which validates the effectiveness of FRSH. Meanwhile, the **Baseline+ GLSTM** embedded in FRSH again improved the average $F1$, OA, and mIoU on both datasets by at least 0.20%, 0.21%, and 0.38%, respectively, which verified that the proposed modules can be combined effectively. Meanwhile, the FRSH module has a small number of parameters, and after embedding it into the network, the number of parameters only increases by 0.11M and the FLOPs increase by 27G. To further demonstrate the advantages of the proposed FRSH module, we replaced the FRSH module in CGGLNet and constructed a variant of CGGLNet using other state-of-the-art fused deep and shallow feature modules for ablation studies. Although the use of the FRSH module increases the computational complexity of the network, its adequate fusion of deep and shallow features through multiple perspectives allows us to achieve the highest mIoU on the Vaihingen dataset, as shown in Table III.

2) Effectiveness of Category-Guided Supervised Transformer: We replace the three CGSTM modules in CGGLNet with GLTM modules to form a new baseline model. Then, the validity of CGSTM is verified by replacing the four GLTM modules one by one. We conducted multiple sets of

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TABLE II
ABLATION STUDY OF DIFFERENT GLOBAL–LOCAL MODULES ON THE VAIHINGEN DATASET

Global-Local Module	Params (M)	FLOPs (GFLOPs)	Mean F1(%)	OA(%)	mIoU (%)
GLTB	**29.72**	**489**	90.61	91.06	83.02
M2SA	34.27	588	90.72	91.14	83.25
CGSTM (Ours)	34.83	605	**91.12**	**91.47**	**83.89**

*The optimal values in each column are in bold.

TABLE III
ABLATION STUDY OF DIFFERENT FEATURE REFINEMENT MODULES ON THE VAIHINGEN DATASET

Feature Refinement Modules	Params (M)	FLOPs (GFLOPs)	Mean F1(%)	OA(%)	mIoU (%)
FRH	35.36	602	90.93	91.20	83.58
MAF	35.33	**594**	90.99	91.22	83.67
FRSH (Ours)	**34.83**	605	**91.12**	**91.47**	**83.89**

*The optimal values in each column are in bold.

experiments on the Vaihingen dataset, and the results of the experiments are shown in Table IV. It is observed that the number of parameters and FLOPs of the network slightly decreases when replacing the first three layers of GLTM one by one, due to the reduction of the $\boldsymbol{Q}$ and $\boldsymbol{K}$ dimensions of the CGSTM compared to the GLTM from 192 to 6. Despite this, the segmentation accuracy continuously improves, which validates the category-guided global contextual modeling effectiveness. However, there is a significant decrease in accuracy when performing category bootstrapping on the deepest features. This is because the downsampling multiplicity of the deepest features is too large, and each point contains extremely rich features, resulting in a serious loss of some distributional information in the spatial dimension. Therefore, it is difficult to use the coarse segmentation results for guided global context fusion. Therefore, we still use the unsupervised multihead transformer for modeling the global context information of the deepest features.

3) Selection of Loss Function Hyperparameter: Loss functions play a key role in machine learning and are essential to guide and improve the model. There are two loss functions in this article, the final segmentation loss and the coarse segmentation loss. To weigh the effects of these two, we introduce the hyperparameter λ, which is used to adjust the weight of the coarse segmentation loss. For experiments on the Vaihingen dataset, we tried λ taking values of 0.2, 0.4, 0.6, 0.8, and 1.0. The results of the experiments are shown in Table V, where the accuracy exhibits a certain degree of improvement as the value of λ increases. This is because the accuracy of the coarse segmentation results input to CGSTM is better ensured as the weights increase, effectively guiding the global context modeling. However, it should be noted that the value of λ cannot be larger than 1.0. Beyond this range, the network will be more inclined to learn good coarse segmentation results from incomplete features and ignore the accuracy of the final segmentation results, leading to a decrease in the overall segmentation accuracy.

4) Selection of the Number of Decoder Channels: To induce a better fusion of features at different scales, we chose to unify the number of feature channels in the encoder. We tried three channel numbers 96, 192, and 288, and show the results of the quantization experiments in the Vaihingen dataset in Table VI. When the number of channels is 96, the network complexity is reduced, but the accuracy is also reduced, showing insufficient feature representation. When the number of channels is 288, the model theoretically has more feature expression, but the test set accuracy decreases due to overfitting the training data. The channel number selection of 192 can achieve a better balance, so the final choice of the decoder channel number is unified as 192.

B. Experimental Results on the ISPRS Vaihingen

In our study, we analyze our approach in comparison with some typical and state-of-the-art semantic segmentation methods, including three classical methods: the FCN8s [12], U-Net [13], and DeepLabv3+ [18]; multiscale attention-based methods: the attentive bilateral contextual network (ABCNet) [50], MANet [21], MAResUNet [20], and global context dependencies aware network (GCDNet) [51]; transformer-based methods: dc-Swin [25], UNetFormer [27], AerialFormer [46], and CMTFNet [28]; and edge information embedding-based approaches: the edge-aware neural network (EaNet) [52] and edge Information guided network (EIGNet) [49], and the SLCNet [24] based on supervised correlated remote modules.

The experimental results of the different methods on the ISPRS Vaihingen dataset are shown in Table VII. As in other experiments, we do not present the correlation accuracy due to the low clutter/background in the dataset. The results in Table VII show that our proposed CGGLNet method achieves the highest scores for mean $F1$, mIoU, and OA. Relative to the suboptimal method, mean $F1$ improves by 0.59%, mIoU improves by 0.96%, and OA improves by 0.48%. In addition, our method achieved the highest values for $F1$ score and IoU for individual categories. In particular, in the small-size target vehicle category, our $F1$ score and IoU reach 90.5% and 82.66%, respectively, which are 1.06% and 1.77% higher than the second-ranked CMTFNet.

To visually compare the segmentation results of different algorithms, we show the segmentation results on the Vaihingen dataset, as shown in Fig. 7, where the size of each image is 1024 × 1024. Classical CNN algorithms, such as FCN8s, U-Net, and DeepLabv3+, are difficult to ensure the completeness of the segmented objects and the accuracy of edge localization because they mainly focus on local features. Recent techniques, such as ABCNet, MAResU-Net, and MANet, improve global context information acquisition by introducing an attention mechanism, but since they all use an LAM, although the computational cost is greatly reduced, the accuracy of their global context information

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TABLE IV
QUANTITATIVE RESULTS USING CGSTM FOR DIFFERENT SCALE FEATURES ON THE VAIHINGEN DATASET (NEW BASELINE = BASELINE + GLTM + ALFE + FRSH)

Method	CGSTM1	CGSTM2	CGSTM3	CGSTM4	Params (M)	FLOPs (GFLOPs)	Mean F_1(%)	OA(%)	mIoU (%)
New Baseline					35.28	618	90.63	90.98	83.07
	√				35.13	608	90.78	91.17	83.33
	√	√			34.98	606	90.99	91.32	83.67
	√	√	√		34.83	605	**91.12**	**91.47**	**83.89**
	√	√	√	√	34.69	605	90.82	91.21	83.38

*The optimal values in each column are in bold.

TABLE V
QUANTITATIVE RESULTS USING DIFFERENT HYPERPARAMETERS FOR THE LOSS FUNCTION ON THE VAIHINGEN DATASET

λ	Mean F_1(%)	OA(%)	mIoU (%)
0.2	90.84	91.11	83.44
0.4	90.97	91.33	83.65
0.6	90.93	91.23	83.58
0.8	91.01	91.38	83.71
1.0	**91.12**	**91.47**	**83.89**

*The optimal values in each column are in bold.

TABLE VI
QUANTITATIVE RESULTS USING DIFFERENT NUMBERS OF DECODER CHANNELS ON THE VAIHINGEN DATASET

Number of decoder channels	Params (M)	FLOPs (GFLOPs)	Mean F_1(%)	OA(%)	mIoU (%)
96	26.63	434	90.88	91.28	83.50
192	34.83	605	**91.12**	**91.47**	**83.89**
288	48.32	888	90.96	91.33	83.63

*The optimal values in each column are in bold.

is sacrificed, resulting in segmentation results that are still incomplete. Transformer-based dc-Swin and UNetFormer utilize transformer to extract global context information, and these implicit global context information acquisitions lead to overlapping pixel semantic content between objects; thus, the edge localization is not sufficiently accurate. EaNet, GCDNet, and EIGNet improve object smoothness of edges between objects but are easily affected by false edges, resulting in incomplete segmentation. CMTFNet uses a multiscale transformer module to extract global context information, which reduces the model complexity and results in the loss of some global features, making the segmentation less complete. SLCNet achieves good results in global context modeling by supervised attention mechanism, but it may result in the overlapping of different categories of features due to the way of weight map generation. In contrast, the proposed method in this article obtains the coarse segmentation results from the multichannel features and performs some fine-tuning on the coarse segmentation results. The pixel features of the same class with high confidence are fully interacted to get intraclass consistent features. The features of different classes are also learned with low weights to make the interclass features more distinguishable.

Comprehensively, it can be seen from the figure that the segmentation effect of CGGLNet is better than other algorithms, especially in the region marked by the red box. For the building classes in the dataset with larger sizes and irregular edges, CGGLNet can segment them more completely, and the edge contours between objects are clearer. In the scene of small-sized target cars, CGGLNet can better localize the edges between dense vehicles, which reflects the complete and accurate segmentation ability for objects of different scales.

C. Experimental Results on the ISPRS Potsdam

As with the ISPRS Vaihingen dataset, we conducted experiments on the larger ISPRS Potsdam dataset. The results in Table VIII show that our proposed CGGLNet method achieves the highest scores in mean $F1$, mIoU, and OA. Relative to the suboptimal method, mean $F1$ improved by 0.43%, mIoU by 0.78%, and OA by 0.46%. In addition, our method achieved the highest values in terms of $F1$ score and IoU for individual categories (except for low vegetation). In particular, in the small-size target vehicle category, our $F1$ score and IoU reach 96.69% and 93.59%, respectively, which are 0.44% and 0.81% higher than the second-ranked EIGNet.

To compare the segmentation results of different algorithms more intuitively, we similarly show the segmentation results of several test sets, as shown in Fig. 8, where each image has a size of 1024 × 1024. We purposely selected four sets of images, which are considered to be somewhat challenging to segment. For example, the red boxes in the first and third sets contain buildings with local textures similar to vegetation and trees. In the red boxes of the fourth group, some buildings appear with local textures similar to impervious surfaces. When coping with the segmentation task of these complex objects, many of the network segmentation results are worse in terms of both completeness and edge localization compared to the ISPRS Vaihingen dataset. However, the segmentation results of our proposed CGGLNet still maintain a high level and significantly outperform other algorithms in general, especially in the regions marked by red boxes. This suggests that guided by categories, our algorithm effectively extracts global contextual information, mining both the more consistent features within the objects as well as effectively capturing the

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TABLE VII
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE ISPRS VAIHINGEN DATASET

Method	Per-Class FI Score (%)/Per-Class loU Score (%)					Mean F_1(%)	OA(%)	mIoU (%)
	Imp.Surf.	Building	Low Veg.	Tree	Car			
FCN8s[12]	91.08/83.63	94.04/88.75	82.51/70.22	88.59/79.52	82.74/70.56	87.79	89.06	78.54
U-Net[13]	91.95/85.11	95.05/90.58	83.85/72.19	89.70/81.33	83.71/71.99	88.85	90.20	80.24
DeepLabv3+[18]	92.27/85.64	95.40/91.21	83.56/71.77	89.22/80.55	85.36/74.46	89.16	90.27	80.73
EaNet[52]	92.27/85.66	95.38/91.18	83.72/72.01	89.64/81.22	86.24/75.80	89.45	90.45	81.17
ABCNet[50]	92.57/86.22	95.43/91.21	83.95/72.33	89.62/81.12	85.40/74.60	89.39	90.42	81.10
MANet[21]	92.42/85.92	95.30/91.02	83.28/71.35	89.55/81.08	87.35/77.54	89.58	90.39	81.38
DC-Swin[25]	92.88/86.71	95.95/92.21	84.20/72.72	89.71/81.34	85.19/74.21	89.59	90.87	81.44
MAResUNet[20]	92.78/86.54	95.71/91.77	84.39/73.00	89.65/81.24	85.23/74.27	89.55	90.77	81.37
UNetFormer[27]	92.59/86.20	95.55/91.49	83.76/72.06	89.57/81.11	88.26/78.99	89.95	90.60	81.97
AerialFormer [46]	93.06/87.02	95.79/91.92	84.12/72.59	89.83/81.54	87.59/77.92	90.08	90.88	82.20
GCDNet[51]	92.62/86.26	95.13/90.72	84.65/73.39	90.10/81.99	89.32/80.71	90.37	90.62	82.61
EIGNet[49]	92.96/86.85	95.84/92.03	84.91/73.77	90.07/81.93	88.92/80.05	90.51	90.99	82.93
CMTFNet[28]	92.90/86.74	95.92/92.16	84.27/72.83	89.85/81.57	89.44/80.89	90.48	90.97	82.84
SLCNet[24]	92.74/86.47	95.70/91.75	84.80/73.62	90.30/82.33	89.08/80.31	90.53	90.81	82.90
Ours	**93.34/87.52**	**96.10/92.51**	**85.32/74.40**	**90.33/82.36**	**90.50/82.66**	**91.12**	**91.47**	**83.89**

*The optimal values in each column are in bold and the sub-optimal values are underlined.

TABLE VIII
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART MODELS ON THE ISPRS POTSDAM DATASET

Method	Per-Class FI Score (%)/Per-Class loU Score (%)					Mean F_1(%)	OA(%)	mIoU (%)
	Imp.Surf.	Building	Low Veg.	Tree	Car			
FCN8s[12]	90.91/83.33	94.66/89.87	85.43/74.57	74.57/76.32	93.84/88.41	90.28	88.59	82.50
U-Net[13]	91.89/85.01	95.77/91.90	85.86/75.22	86.87/76.79	94.58/89.71	90.99	89.38	83.73
DeepLabv3+[18]	91.90/85.02	96.03/92.37	86.38/76.02	87.52/77.81	94.73/89.99	91.31	89.83	84.24
EaNet[52]	92.36/85.81	96.08/92.46	87.03/77.05	88.68/79.67	95.36/91.13	91.90	90.42	85.22
ABCNet[50]	92.38/85.84	96.63/93.48	86.74/76.59	88.29/79.04	95.21/90.86	91.85	90.40	85.16
MANet[21]	92.95/86.83	96.67/93.56	86.74/76.59	88.35/79.13	95.54/91.47	92.04	90.70	85.51
DC-Swin[25]	93.07/87.04	96.68/93.59	87.42/77.64	88.63/79.59	95.51/91.40	92.26	90.84	85.86
MAResUNet[20]	92.90/86.74	96.22/92.72	86.65/76.45	88.41/79.23	95.78/91.90	91.99	90.56	85.41
UNetFormer[27]	93.23/87.32	96.67/93.57	86.82/76.71	88.13/78.77	96.02/92.35	92.17	90.70	85.75
AerialFormer [46]	93.23/87.34	97.08/94.37	**87.81/78.27**	89.12/80.39	95.43/91.27	92.53	91.12	86.33
GCDNet[51]	92.95/86.83	96.67/93.55	87.31/77.49	88.35/79.14	96.21/92.70	92.30	90.66	85.94
EIGNet[49]	93.09/87.08	96.82/93.83	87.30/77.45	88.64/79.60	96.25/92.78	92.42	90.91	86.14
CMTFNet[28]	93.28/87.41	97.07/94.32	87.08/77.13	88.42/79.25	95.93/92.17	92.36	90.92	86.05
SLCNet[24]	93.21/87.28	96.99/94.17	87.20/77.41	88.99/80.04	95.60/91.57	92.40	90.94	86.10
Ours	**93.84/88.40**	**97.44/95.02**	87.66/78.03	**89.18/80.48**	**96.69/93.59**	**92.96**	**91.58**	**87.11**

*The optimal values in each column are in bold and the sub-optimal values are underlined.

dissimilar features among the objects. The ALFEM multiscale local feature adaptive extraction allows CGGLNet to outperform when dealing with both large- and small-scale objects. In addition, FRSH effectively fuses deep semantic features with shallow features, which makes the final segmentation results have more accurate localization at the edges. Comprehensively, our proposed CGGLNet successfully achieves complete segmentation of multiscale objects in semantic segmentation of remote sensing images and shows high accuracy in edge localization. Compared with other algorithms, our method achieves better segmentation results when dealing with some challenging localized images.

D. Model Parameters and Computation Complexity Analysis

We fed input data of size $3 \times 1024 \times 1024$ into all models and also evaluated the computational complexity and parameter sizes of various methods in the same runtime environment, and the results are shown in Table IX. From Table IX, it can be seen that the number of model parameters of CGGLNet is at a medium level compared to other methods,

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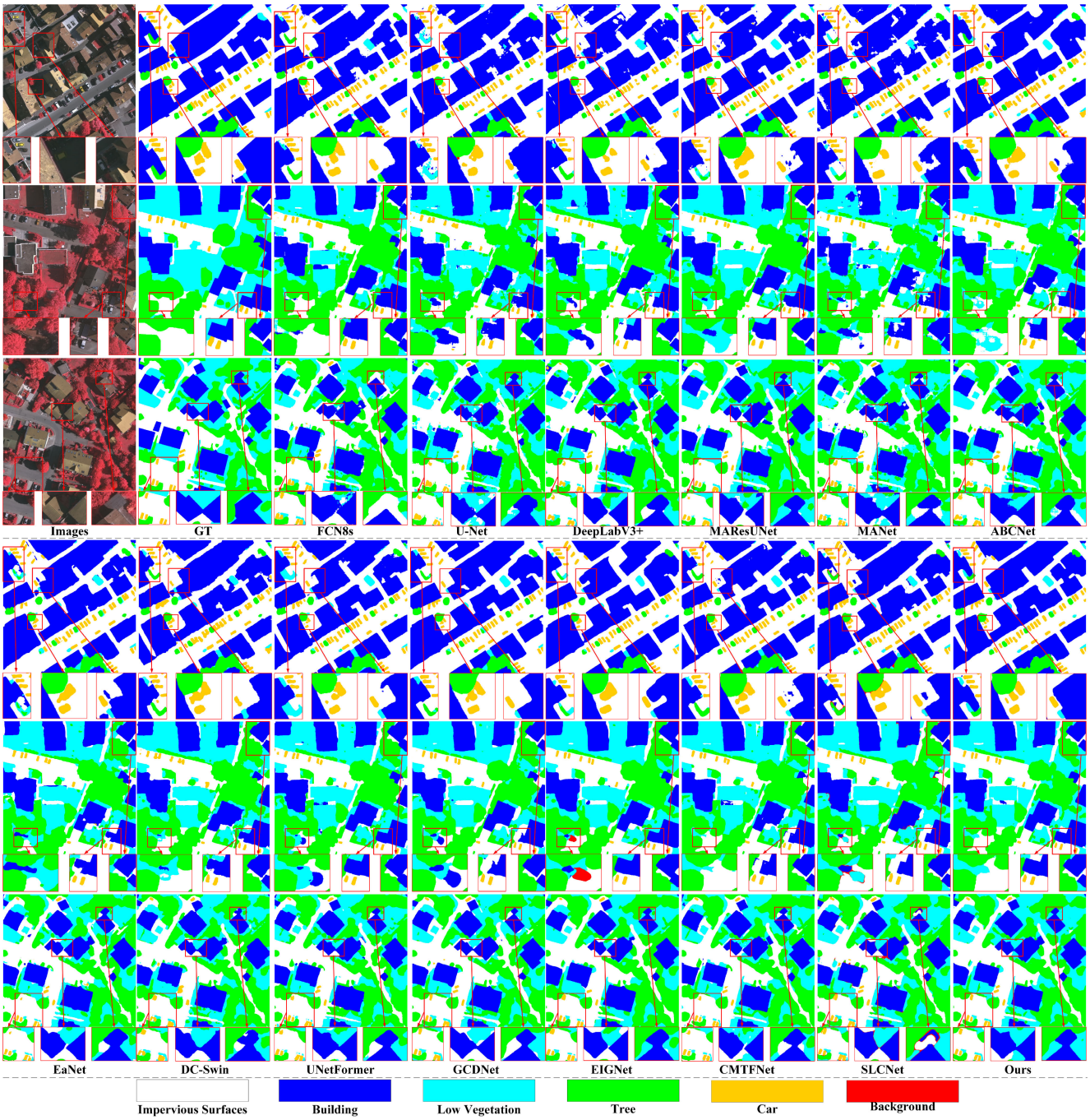


Fig. 7. Qualitative comparison of the visualization results of our method with other methods on the ISPRS Vaihingen dataset. The red-boxed parts of the figure show some localizations where segmentation is significantly improved.

but the computational complexity of the models is higher. This is mainly due to the introduction of the transformer, which is more computationally intensive. Overall, although CGGLNet is in the middle of the range in terms of computational complexity, it shows superior performance.

VI. Discussion

A. Visualization of the Category-Guided Weight Map

To demonstrate the mechanism of CGSTM more intuitively, we visualize the weight maps it generates. As shown in Fig. 9, the multihead weight maps of the shallowest features when performing global context modeling are demonstrated. From left to right, the input image and the semantic segmentation truth map, the vertical direction weight map generated by the classical multihead transformer module, the vertical direction weight map generated by CGSTM, the vertical direction truth weight map for each category, as well as the weight map generated by the above modules in the horizontal direction, and the weight map truth in the horizontal direction are shown. Taking the feature maps in the vertical direction as an example, CGSTM generated weight maps in the $W/4 \times H/4 \times H/4$ dimension for each head. For presentation purposes, the

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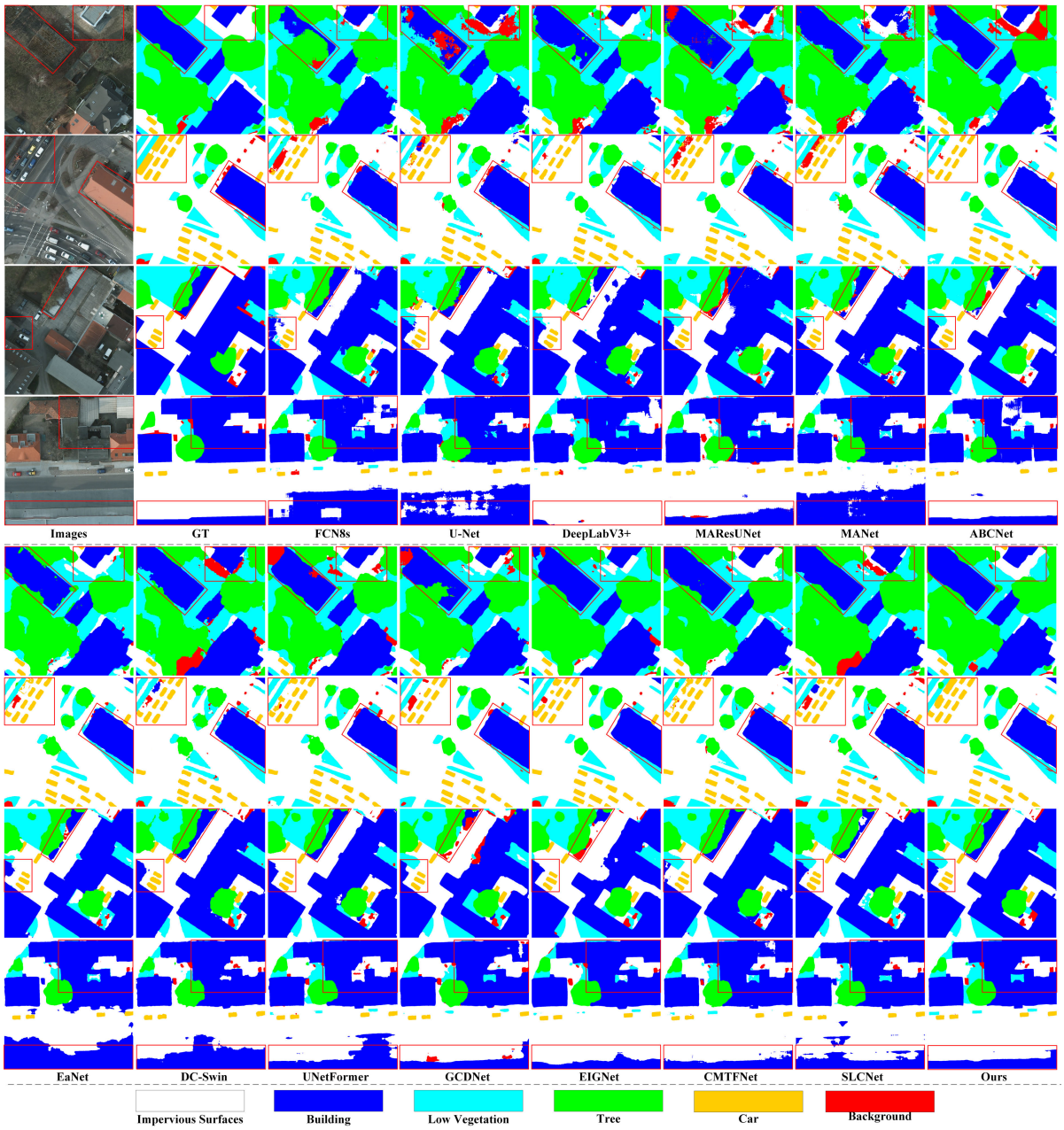


Fig. 8. Qualitative comparison of the visualization results of our method with other methods on the ISPRS Potsdam dataset. The red-boxed parts of the figure show some localizations where segmentation is significantly improved.

dimension $W/4$ is averaged to obtain a weight map for each head.

From the figure, it can be seen that the unsupervised multihead weight map has no practical categorical significance and prefers to perform grand unified feature interactions. In contrast, each head of our proposed CGSTM module assigns a high weight to a separate category, much like the truth-value weight map we expect to learn. This allows features of pixels belonging to that class to interact sufficiently, while features of pixels not belonging to that class are assigned lower weights for interaction, thus mining features with more intraclass consistency and interclass differentiation.

At the same time, SLCNet is directly supervising the generated weight maps, which may introduce some problems. First, the computational cost of the model can increase significantly. Compared to the computation of only the feature map constraining $N \times H \times W$, the weight map constraining multiple $W/4 \times H/4 \times H/4$ will make the computation of the network surge greatly, which is not conducive to larger scale training with the same computational resources. Second, directly supervising the weight maps and forcing each head to learn the true value weight map is detrimental to the training of the network. Neural networks use features to classify each pixel point, and many misclassified pixel points have features that approximate the features of the pixel points in the category they are misclassified in. Forcing the features of these misclassified pixel points to interact with the features of the pixel points that rightfully belong

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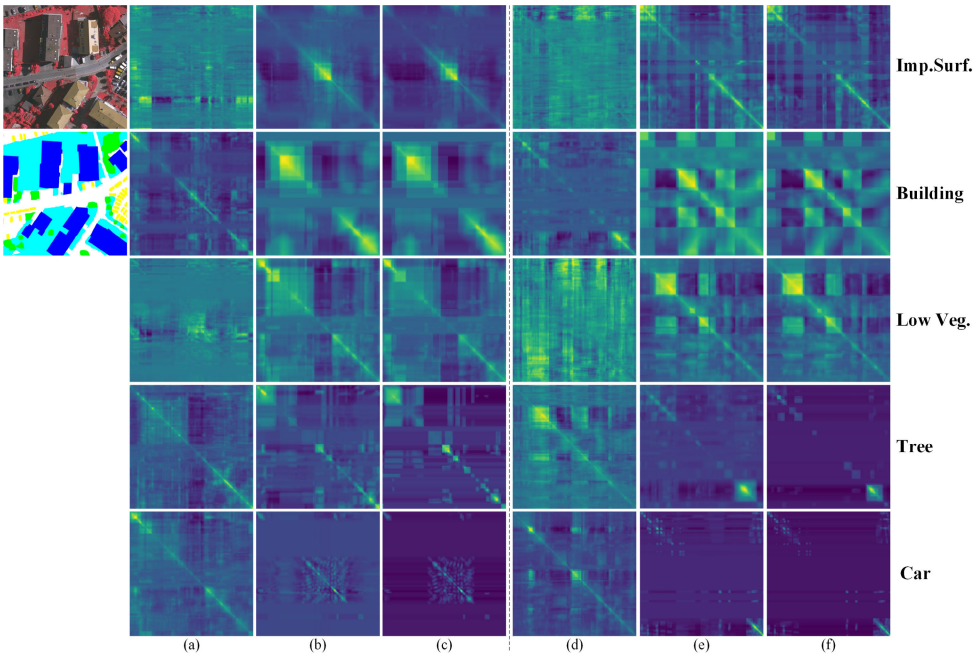


Fig. 9. Weight maps of each head for CGSTM and classical multihead converter in ISPRS Vaihingen results. (a) and (d) Weight diagrams of the classical multihead transformer in vertical and horizontal directions. (b) and (e) Weight maps of the CGSTM in vertical and horizontal directions. (c) and (f) Theoretical weight maps for each category in vertical and horizontal directions. Each row represents five categories: impervious surfaces, buildings, low vegetation, trees, and cars.

TABLE IX
COMPARISON OF THE COMPLEXITY OF DIFFERENT NETWORKS

Method	Params(Mb)	FLOPs(Gbps)
FCN8s[12]	18.64	190.37
U-Net[13]	28.29	546.59
DeepLabv3+[18]	40.35	186.79
EaNet[52]	102.57	746.29
ABCNet[50]	13.96	62.34
MANet[21]	35.68	310.47
DC-Swin[25]	66.95	276.32
MAResUNet[20]	26.28	140.04
UNetFormer[27]	**11.68**	**46.83**
AerialFormer [46]	67.51	379.31
GCDNet[51]	58.71	386.54
EIGNet[49]	34.48	318.80
CMTFNet[28]	30.07	130.00
SLCNet[24]	187.15	2025.00
Ours	34.83	605.00

*The optimal values in each column are in bold.

to the category can adversely affect the learning of the network.

For the CGSTM module, we used a residual structure to estimate the coarse segmentation result for the multichannel features, which was generated from the perspective of the extracted features. At the same time, we recognize that this result may be inaccurate and therefore use the residual structure in the generation to make it somewhat learnable so that the network can gradually correct some misclassified points in the coarse segmentation result.

B. Visualization of Heatmaps for GLSTM

In this section, we will use Grad-class activation mapping (CAM) [53] to visualize the responses of the feature maps before and after GLSTM for different categories of objects, further demonstrating the superiority of GLSTM for global and local contextual information modeling. As shown in Fig. 10, we show the responses of the features before processing when facing different categories, after global context modeling by multihead transformer, and after global and local context modeling by GLSTM on different categories (the response maps are obtained by fusing the four-scale feature maps). It can be found that there is some degree of confusion and noise in the responses of the features on different categories before processing. The features after processing by multihead transformer have a great improvement in completeness but still have a certain degree of noisy response. The features after GLSTM processing focus on the inside of the categories with higher response values, especially for the two categories of buildings and trees, which are objects with large-scale variations, GLSTM presents better multiscale response characteristics compared to the features after multihead transformer processing.

C. Coarse Segmentation Result Generation Strategy

The CGGLNet proposed in this article is to estimate and constrain the coarse segmentation results for each of the three layers of different size maps to guide the global context information modeling. In addition to this strategy, we also

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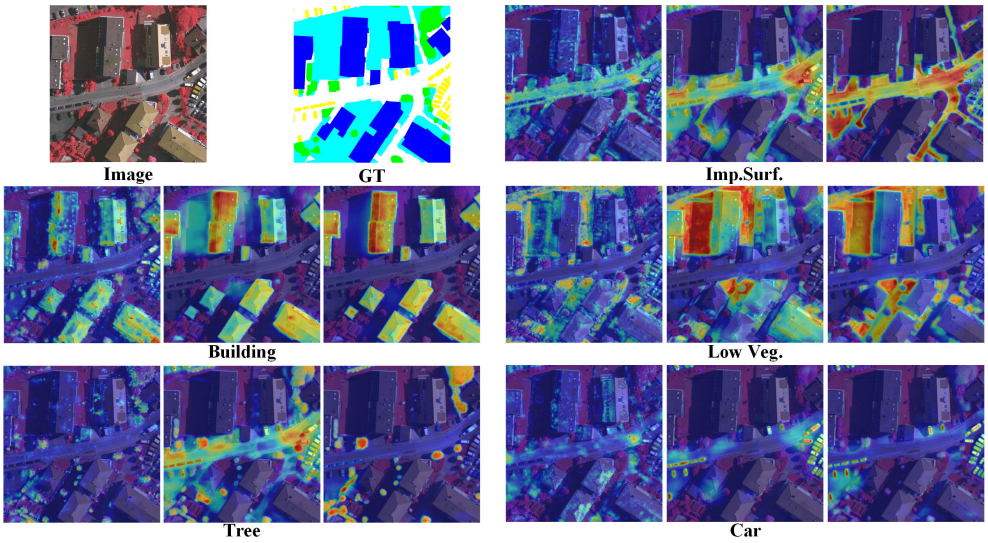


Fig. 10. Heatmap of the responses of features of different categories before and after GLSTM processing. The left side shows the responses of the features facing different categories of objects before input, while the center and right sides show the responses of the features facing different categories of objects after multihead transformer processing and GLSTM processing, respectively.

adopt an auxiliary network to fuse multiscale feature maps to generate more accurate coarse segmentation results to guide the global contextual information modeling of multichannel features of different scales. Experimentally, it is found that the segmentation is not as good as the strategy adopted by CGGLNet. The coarse segmentation results obtained from the fused features are too fine, and some similar pixels may not have obvious similarity in the current layer, forcing these pixels to interact with each other may have a negative impact.

VII. Conclusion

In this article, we propose a novel network framework for remote sensing image semantic segmentation task called CGGLNet. To better model the global and local contextual information of each category for more complete segmentation of multiscale objects, we design a GLSTM containing a CGSTM and an ALFEM. In addition, we introduce the FRSH to reduce the semantic difference between shallow and deep features and promote their better fusion. Through extensive ablation experiments, comparison tests, and discussions on two high-resolution remote sensing datasets (ISPRS Vaihingen and ISPRS Potsdam), we validate the effectiveness of the proposed method. In future research, we will further explore ways to reduce the computational cost of the category-guided global contextual information module to apply the method more widely.

Acknowledgment

The authors are grateful to the creators of the comparison algorithm for generously sharing their code. They also thank the editors and reviewers for their valuable comments.

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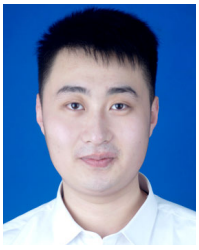
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